

# Quadratics

Solve it, complete the square, read the discriminant. Every quadratic question is one of these three — and the graph ties them together.



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## THE TOPIC ON ONE MAP

### Quadratics

#### 01 — Solve: calculator first

find the roots

surd form

word problem

#### 02 — Complete the square

$a(x + b)^2 + c$

turning point

line of symmetry

#### 03 — The discriminant

two distinct

equal roots

no real roots

#### 04 — Hidden quadratics

$x^4 - 13x^2 + 36$

$2^{2x}$

$\sqrt{x}$

#### 05 — Graphs and models

sketch

maximum height

# 01 Solve: Calculator First, Then Show It

Get the roots on the calculator first. Clean roots → factorise. Messy roots → formula or completing the square.

## WHEN YOU SEE → YOU DO

**“solve”, “find the roots”**

→ calculator roots, then the working

**“using algebra” / “calculator solutions not acceptable”**

→ show factorising or the formula

**“exact” / “surd form”**

→ keep the  $\sqrt{\phantom{x}}$

**a word problem (perimeter, area)**

→ form two equations → substitute → quadratic

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (\text{brackets around a negative } b)$$

## WORKED MODEL · JUNE 2023

Rectangle: perimeter 350, area 7350,  $x > y$ .

1  $x + y = 175, xy = 7350$

two equations

2  $x(175 - x) = 7350 \Rightarrow x^2 - 175x + 7350 = 0$

substitute

3  $(x - 105)(x - 70) = 0$

factorise

**ANSWER**  $x = 105, y = 70$

## IDEA 01

**Practise: Solve**

My questions first, then selected practice, then real past papers.

**Q1** **DR ESLAM** Calculator, then show it

P1 · TOPIC 2 3 marks

Solve (a)  $x^2 - 6x + 5 = 0$  (b)  $2x^2 + 3x - 4 = 0$  (exact)

**SELECTED PRACTICE**

**Q2** **PRACTICE** Practice · Easy

P1 · TOPIC 2 3 marks

1 Expand and simplify

(i)  $(x + 4)(2x - 3)$

(ii)  $(3x - 4)(3x + 4)$

(iii)  $(2x + 1)^2$

(3 marks)

**Q3****PRACTICE**

Practice · Easy

**P1 · TOPIC 2 3 marks****4** Solve

(i)  $x^2 + 8x - 9 = 0$

(ii)  $3x^2 - 13x + 4 = 0$

(iii)  $4x^2 - 6x - 5 = 0$

**(3 marks)**

**Q4****PRACTICE**

Practice · Very Hard

**P1 · TOPIC 2****5 marks****12 (a)** Factorise  $x^2 + 6x + 9$ **(1 mark)****(b)** Factorise  $x^2 + 6xy + 9y^2$ **(2 marks)****(c)** Find a relationship between  $x$  and  $y$  such that  $x^2 + 6xy + 9y^2 = 0$ **(2 marks)**

## PAST PAPERS

**Q5** PAST PAPER Pearson IAL WMA11 · June 2023 · Q2

P1 · TOPIC 2 6 marks

2. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A rectangular sports pitch has length  $x$  metres and width  $y$  metres, where  $x > y$

Given that the perimeter of the pitch is 350 m,

(a) write down an equation linking  $x$  and  $y$  (1)

Given also that the area of the pitch is  $7350\text{ m}^2$

(b) write down a second equation linking  $x$  and  $y$  (1)

(c) hence find the value of  $x$  and the value of  $y$  (4)

(Total for Question 2 is 6 marks)

## 02 Completing the Square and the Turning Point

The same two-step structure from O-Level: list  $a$ ,  $b$ ,  $c$ ,  $\frac{b}{2a}$ , then  $a\left(x + \frac{b}{2a}\right)^2 - a\left(\frac{b}{2a}\right)^2 + c$ .

### WHEN YOU SEE → YOU DO

“in the form  $a(x + b)^2 + c$ ”, “ $a + b(x + c)^2$ ”

→ complete the square

“minimum/maximum point”, “turning point”

→ flip the inside, keep the outside

“minimum value of  $f(x)$ ”

→ the number outside the bracket

“line of symmetry”

→  $x$  = the flipped inside number

“use inequalities to define  $R$ ”

→ one inequality per boundary:  $x \geq 0$ ,  
 $y \leq \text{curve}$ ,  $y \geq \text{line}$

### WORKED MODEL · OCTOBER 2023

$$2x^2 - 12x + 14$$

1  $a = 2$ ,  $b = -12$ ,  $c = 14$ ,  $\frac{b}{2a} = -3$

list

2  $2(x - 3)^2 - 2(9) + 14$

structure

ANSWER  $2(x - 3)^2 - 4$ ,  $P(3, -4)$

× **TRAP** A negative  $a$  flips the curve:  $-6x^2 + \dots$  has a **maximum**.

## IDEA 02

**Practise: Completing the Square**

My questions first, then selected practice, then real past papers.

**Q6** DR ESLAM Turning point

P1 · TOPIC 2 3 marks

Write  $x^2 - 6x + 5$  in the form  $(x + p)^2 + q$  and state the turning point of  $y = x^2 - 6x + 5$ .

**SELECTED PRACTICE****Q7** PRACTICE Practice · Easy

P1 · TOPIC 2 3 marks

3 Complete the square for

(i)  $x^2 + 8x - 4$

(ii)  $2x^2 + 12x - 5$

(iii)  $5x^2 - 3x + 2$

**(3 marks)**

**Q8****PRACTICE**

Practice · Hard

**P1 · TOPIC 2****6 marks**

- 1 (a) Write the quadratic function  $y = 4x^2 + 8x - 5$  in the form  $y = a(x + b)^2 + c$  where  $a$ ,  $b$  and  $c$  are integers to be found.

**(2 marks)**

- (b) Write down the minimum point on the graph of  $y = 4x^2 + 8x - 5$ .

**(1 mark)**

- (c) Sketch the graph of  $y = 4x^2 + 8x - 5$ , clearly labelling the minimum point and any point where the graph intersects the coordinate axes.

**(3 marks)**

**Q9****PRACTICE**

Practice · Very Hard

**P1 · TOPIC 2 4 marks**

7 (a) Show that the equation  $ax^2 + bx + c = 0$  can be written in the form

$$a\left(x + \frac{b}{2a}\right)^2 - \frac{(b^2 - 4ac)}{4a} = 0$$

**(2 marks)**

(b) Hence show that  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ .

**(2 marks)**

## PAST PAPERS

**Q10** PAST PAPER Pearson IAL WMA11 · January 2022 · Q2

P1 · TOPIC 2 6 marks

2.  $f(x) = 11 - 4x - 2x^2$

- (a) Express
- $f(x)$
- in the form

$$a + b(x + c)^2$$

where  $a$ ,  $b$  and  $c$  are integers to be found.

(3)

- (b) Sketch the graph of the curve
- $C$
- with equation
- $y = f(x)$
- , showing clearly the coordinates of the point where the curve crosses the
- $y$
- axis.

(2)

- (c) Write down the equation of the line of symmetry of
- $C$
- .

(1)

**(Total 6 marks)**

**Q11** PAST PAPER Pearson IAL WMA11 · June 2023 · Q3

P1 · TOPIC 2 6 marks

3. (a) Express  $3x^2 + 12x + 13$  in the form

$$a(x + b)^2 + c$$

where  $a$ ,  $b$  and  $c$  are integers to be found.

(3)

- (b) Hence sketch the curve with equation  $y = 3x^2 + 12x + 13$

On your sketch show clearly

- the coordinates of the  $y$  intercept
- the coordinates of the turning point of the curve

(3)

(Total for Question 3 is 6 marks)

**Q12****PAST PAPER**

Pearson IAL WMA11 · October 2023 · Q11

**P1 · TOPIC 2 10 marks**

11.

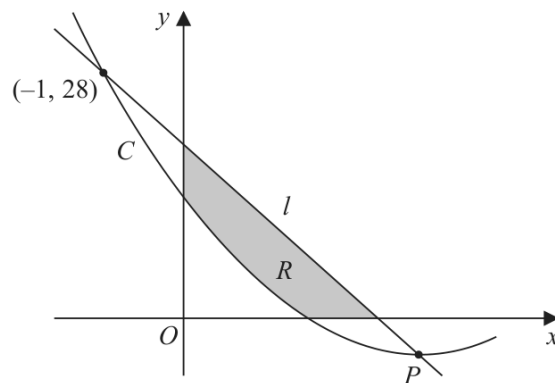
**Figure 5**

Figure 5 shows part of the curve  $C$  with equation  $y = f(x)$  where

$$f(x) = 2x^2 - 12x + 14$$

- (a) Write  $2x^2 - 12x + 14$  in the form

$$a(x + b)^2 + c$$

where  $a$ ,  $b$  and  $c$  are constants to be found.

**(3)**

Given that  $C$  has a minimum at the point  $P$

- (b) state the coordinates of  $P$

**(1)**

The line  $l$  intersects  $C$  at  $(-1, 28)$  and at  $P$  as shown in Figure 5.

- (c) Find the equation of  $l$  giving your answer in the form  $y = mx + c$  where  $m$  and  $c$  are constants to be found.

**(3)**

The finite region  $R$ , shown shaded in Figure 5, is bounded by the  $x$ -axis,  $l$ , the  $y$ -axis, and  $C$ .

- (d) Use inequalities to define the region  $R$ .

**(3)**

**(Total for Question 11 is 10 marks)**

5.

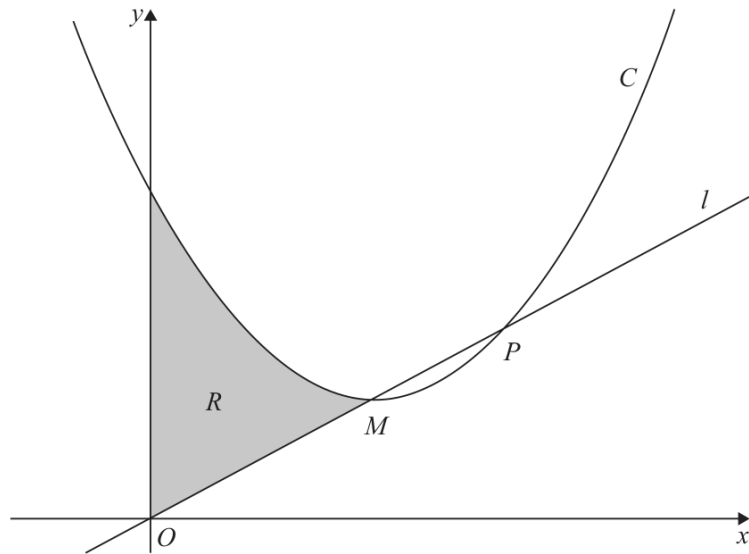


Figure 1

**In this question you must show all stages of your working.  
Solutions relying on calculator technology are not acceptable.**

- (a) Express  $2x^2 - 16x + 50$  in the form

$$a(x + b)^2 + c$$

where  $a$ ,  $b$  and  $c$  are constants to be found.

(3)

Figure 1 shows a sketch of the curve  $C$  with equation  $y = 2x^2 - 16x + 50$

Given that in Figure 1

- $M$  is the minimum point on  $C$
- line  $l$  passes through the origin and intersects  $C$  at the points  $M$  and  $P$

- (b) find, using algebra and showing your working, the  $x$  coordinate of  $P$ .

(5)

The region  $R$  is shown shaded in Figure 1.

- (c) Use inequalities to fully define  $R$ .

(2)

**(Total for Question 5 is 10 marks)**

**Q14** PAST PAPER Pearson IAL WMA11 · June 2026 · Q5

P1 · TOPIC 2 7 marks

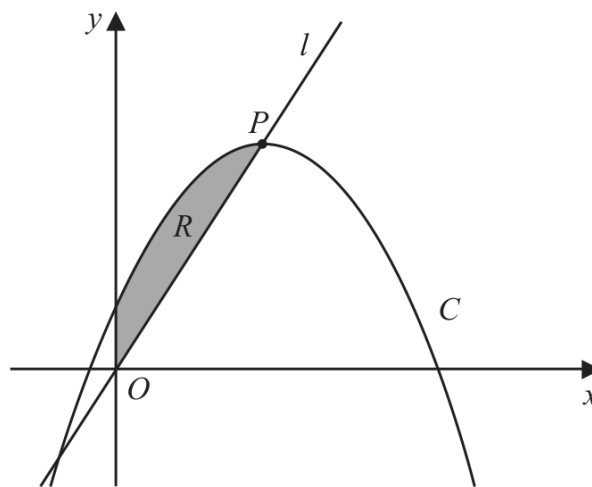
5. The curve  $C$  has equation

$$y = -2x^2 + 12x + 7$$

- (a) Write  $-2x^2 + 12x + 7$  in the form  $a(x + b)^2 + c$ , where  $a$ ,  $b$  and  $c$  are constants. (3)

The point  $P$  is the maximum point on  $C$ .

- (b) Find the coordinates of  $P$ . (2)



**Figure 1**

Figure 1 shows a sketch of the curve  $C$  and a line  $l$ .

The line  $l$  passes through the point  $P$  and the origin  $O$ .

The region  $R$ , shown shaded in Figure 1, is bounded by  $C$ ,  $l$  and the  $y$ -axis.

- (c) Use inequalities to define  $R$ . (2)

**Question 5 continued**

**(Total for Question 5 is 7 marks)**

## 03 The Discriminant $b^2 - 4ac$

It counts the roots without solving. Most exam questions give the number of roots and ask for  $k$ .

### WHEN YOU SEE → YOU DO

“two distinct real roots” →  $b^2 - 4ac > 0$

“equal roots”, “one root”, “touches”, “meet at a single point” →  $b^2 - 4ac = 0$

“no real roots”, “does not cross” →  $b^2 - 4ac < 0$

“real roots” →  $b^2 - 4ac \geq 0$

two curves meet → curve = curve → one quadratic → its discriminant

1

**Rearrange to  $ax^2 + bx + c = 0$**

Read  $a$ ,  $b$ ,  $c$  — they may contain  $k$ .

2

**Write the inequality**

$b^2 - 4ac >, =, \text{ or } < 0$ .

3

**Solve the quadratic inequality in  $k$**

Critical values, sketch, choose inside or outside.

### WORKED MODEL · JANUARY 2023

$kx^2 + 6kx + 5 = 0$  has no real roots.

1  $36k^2 - 20k < 0$

$b^2 - 4ac < 0$

2  $4k(9k - 5) < 0$ , critical values  $0, \frac{5}{9}$

factorise

ANSWER  $0 < k < \frac{5}{9}$

## IDEA 03

**Practise: The Discriminant**

My questions first, then selected practice, then real past papers.

**Q15** DR ESLAM Equal roots

P1 · TOPIC 2 2 marks

$x^2 + kx + 4 = 0$  has equal roots. Find  $k$ .

$k =$  \_\_\_\_\_

**Q16** DR ESLAM No real roots

P1 · TOPIC 2 3 marks

Find the values of  $k$  for which  $x^2 + 4x + k = 0$  has no real roots.

## SELECTED PRACTICE

**Q17****PRACTICE**

Practice · Easy

P1 · TOPIC 2

**3 marks**

5 Write down the value of the discriminant of

(i)  $x^2 - 3x + 4$

(ii)  $4x + 3 - 2x^2$

(iii)  $5 - 8x + 2x^2$

**(3 marks)****Q18****PRACTICE**

Practice · Medium

P1 · TOPIC 2

**2 marks**

7 The equation  $y = x^2 + px + q$  has no real roots. Show that  $p^2 < 4q$ .

**(2 marks)**

**Q19****PRACTICE**

Practice · Very Hard

**P1 · TOPIC 2****5 marks**

**8 (a)** The function  $f(x)$  is defined by  $f(x) = (k-1)x^2 - (k-2)x - 2k$ ,  $x \in \mathbb{R}$ .

The function  $g(x)$  is defined by  $g(x) = (k-1)x^2 - 3kx + k + 1$ ,  $x \in \mathbb{R}$ .

$k$  is a non-zero constant and  $k \neq 1$ .

The graphs of  $y = f(x)$  and  $y = g(x)$  intersect once.

Find the  $x$ -coordinate of the intersection, giving your answer in terms of  $k$ .

**(3 marks)**

**(b)** In the case when  $k = 3$ , find the coordinates of the point of intersection of  $y = f(x)$  and  $y = g(x)$ .

**(2 marks)**

## PAST PAPERS

**Q20** **PAST PAPER** Pearson IAL WMA11 · January 2023 · Q4**P1 · TOPIC 2** **4 marks**

4. Given that the equation

$$kx^2 + 6kx + 5 = 0 \quad \text{where } k \text{ is a non zero constant}$$

has no real roots, find the range of possible values for  $k$ .**(4)****(Total for Question 4 is 4 marks)**

**Q21****PAST PAPER**

Pearson IAL WMA11 · October 2023 · Q6

**P1 · TOPIC 2 6 marks**

6. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

The equation

$$4(p - 2x) = \frac{12 + 15p}{x + p} \quad x \neq -p$$

where  $p$  is a constant, has two distinct real roots.

(a) Show that

$$3p^2 - 10p - 8 > 0 \quad (3)$$

(b) Hence, using algebra, find the range of possible values of  $p$  (3)

(Total for Question 6 is 6 marks)

**Q22** PAST PAPER Pearson IAL WMA11 · June 2024 · Q4

P1 · TOPIC 2 6 marks

4. The curve  $C_1$  has equation

$$y = x^2 + kx - 9$$

and the curve  $C_2$  has equation

$$y = -3x^2 - 5x + k$$

where  $k$  is a constant.

Given that  $C_1$  and  $C_2$  meet at a single point  $P$

(a) show that

$$k^2 + 26k + 169 = 0 \quad (3)$$

(b) Hence find the coordinates of  $P$  (3)

(Total for Question 4 is 6 marks)

**Q23****PAST PAPER**

Pearson IAL WMA11 · June 2025 · Q10

**P1 · TOPIC 2 7 marks****10.**

**In this question you must show all stages of your working.  
Solutions relying on calculator technology are not acceptable.**

$$(k-1)x^6 + 4x^3 + (k-4) = 0 \quad \text{where } k \text{ is a constant}$$

- (a) Find the exact solutions to the given equation for  $k = 4.5$  **(3)**

- (b) Find the set of possible values of  $k$  for which the given equation has no real roots. **(4)**

**(Total for Question 10 is 7 marks)**

## 04 Hidden Quadratics: Substitute

Look for a power that is double another:  $x^4$  and  $x^2$ ,  $x^{\frac{2}{3}}$  and  $x^{\frac{1}{3}}$ ,  $x$  and  $\sqrt{x}$ ,  $4^x$  and  $2^x$ .

### WHEN YOU SEE → YOU DO

$$x^4 - 13x^2 + 36$$

$$\longrightarrow u = x^2$$

$$x - 6x^{\frac{1}{2}} + 4, 5\sqrt{x} + 3 = 2x$$

$$\longrightarrow u = \sqrt{x}$$

$$2^{2x} + 64 = 20(2^x)$$

$$\longrightarrow u = 2^x$$

$$\sqrt{x} \text{ or } 2^x \text{ negative}$$

$$\longrightarrow \text{reject it}$$

### WORKED MODEL · OCTOBER 2019

$$x - 6x^{\frac{1}{2}} + 4 = 0$$

$$1 \quad u^2 - 6u + 4 = 0, \quad u = x^{\frac{1}{2}}$$

substitute

$$2 \quad u = 3 \pm \sqrt{5}$$

solve

$$\text{ANSWER} \quad x = (3 \pm \sqrt{5})^2 = 14 \pm 6\sqrt{5}$$

## IDEA 04

**Practise: Hidden Quadratics**

My questions first, then selected practice, then real past papers.

**Q24** **DR ESLAM** Substitute  $u = x^2$ 

P1 · TOPIC 2 3 marks

Solve  $x^4 - 5x^2 + 4 = 0$ .

$x =$  \_\_\_\_\_

**SELECTED PRACTICE****Q25** **PRACTICE** Practice · Medium

P1 · TOPIC 2 3 marks

10 Solve the equation  $x^4 - 13x^2 + 36 = 0$ .

(3 marks)

**Q26****PRACTICE**

Practice · Hard

**P1 · TOPIC 2 9 marks**

9 (a) Solve the equation  $5\sqrt{x} + 3 = 2x$ .

**(3 marks)**

(b) Solve  $x^{\frac{2}{3}} + 2x^{\frac{1}{3}} = 8$ .

**(3 marks)**

(c) Solve the equation  $2^{2x} + 64 = 20(2^x)$ .

**(3 marks)**

**Q27****PRACTICE**

Practice · Very Hard

**P1 · TOPIC 2 6 marks**

9 (a) Solve the equation  $8\sqrt{x} = 48 - x$ .

**(3 marks)**

(b) Solve the equation  $2^{4x} + 64 = 20(2^{2x})$ .

**(3 marks)**

## PAST PAPERS

**Q28** **PAST PAPER** Pearson IAL WMA11 · October 2019 · Q8**P1 · TOPIC 2** **5 marks**

8. Solve, using algebra, the equation

$$x - 6x^{\frac{1}{2}} + 4 = 0$$

Fully simplify your answers, writing them in the form  $a + b\sqrt{c}$ , where  $a$ ,  $b$  and  $c$  are integers to be found.

**(5)****(Total 5 marks)**

**Q29****PAST PAPER**

Pearson IAL WMA11 · January 2020 · Q5

**P1 · TOPIC 2 7 marks**

5. (a) Find, using algebra, all solutions of

$$20x^3 - 50x^2 - 30x = 0 \quad (3)$$

- (b) Hence find all real solutions of

$$20(y+3)^{\frac{3}{2}} - 50(y+3) - 30(y+3)^{\frac{1}{2}} = 0 \quad (4)$$

**(Total 7 marks)**

**Q30****PAST PAPER**

Pearson IAL WMA11 · October 2025 · Q4

**P1 · TOPIC 2 7 marks**

4. **In this question you must show all stages of your working.  
Solutions relying on calculator technology are not acceptable.**

- (i) Using the laws of indices, solve

$$3^{2p-5} = \frac{1}{9}\sqrt{27}$$

**(3)**

- (ii) Find the real roots of the equation

$$x^2 + 1 = \frac{36}{x^2 - 4}$$

**(4)****(Total for Question 4 is 7 marks)**

## 05 Graphs and Models

A sketch needs three things: shape, the axis crossings, the turning point. Models ask you to read those same points in context.

### WHEN YOU SEE → YOU DO

“sketch ... showing where it crosses the axes”

→ shape ( $a$ ),  $y$ -intercept, roots, turning point

“ $y = f(x + 2)$ ”, “ $y = f(x) + 6$ ”

→ move 2 left · move 6 up

“height of the cliff”, “when does it hit the sea”

→  $t = 0$  · the positive root

“maximum height”, “is the bridge wide enough?”

→ turning point · distance between the roots

“find the points of intersection”

→ curve = line → solve

### WORKED MODEL · STONE FROM A CLIFF

$$h = 24 + 2t - 0.5t^2$$

1  $t = 0$ : cliff 24 m

start

2  $-\frac{1}{2}(t - 2)^2 + 26$

complete the square

3  $0 = 24 + 2t - 0.5t^2 \Rightarrow t = 2 + 2\sqrt{13}$

positive root

**ANSWER** max 26 m; hits the sea at 9.21 s

IDEA 05

## Practise: Graphs and Models

My questions first, then selected practice, then real past papers.

**Q31** DR ESLAM Sketch

P1 · TOPIC 2 3 marks

Sketch  $y = x^2 - 6x + 5$ , showing the axis crossings and the turning point.

## SELECTED PRACTICE

**Q32**

PRACTICE

Practice · Easy

P1 · TOPIC 2 6 marks

6 (a) (i) Write down the  $y$ -axis intercept on the graph of  $y = 2x^2 + 5x - 3$ .

(ii) Find the roots of  $y = 2x^2 + 5x - 3$ .

**(3 marks)**

(b) Sketch the graph of  $y = 2x^2 + 5x - 3$ , labelling all points where the graph crosses the coordinate axes.

**(3 marks)****Q33**

PRACTICE

Practice · Hard

P1 · TOPIC 2 3 marks

6 Sketch the graph of  $y = 12x^2 - 5x - 72$ , labelling any points where the graph crosses the coordinate axes. (You do not need to label the turning point.)

**(3 marks)**

**Q34****PRACTICE**

Practice · Very Hard

**P1 · TOPIC 2 7 marks**

- 11 (a)** A stone is thrown vertically upwards from the top of a cliff. The path of the stone is modelled by the quadratic function  $h(t) = 52 + 3t - 0.5t^2$ ,  $t \geq 0$ , where  $h$  is the height, in meters, of the stone above the sea and  $t$  is the time in seconds since the stone was thrown.

Write down the height of the cliff from which the stone was thrown.

**(1 mark)**

- (b)** Find the maximum height the stone reaches above the sea.

**(2 marks)**

- (c)** How long does it take for the stone to hit the sea?

**(2 marks)**

- (d)** How long does the stone stay above it's starting height for?

**(2 marks)**

## PAST PAPERS

Q35 PAST PAPER Pearson IAL WMA11 · January 2023 · Q8

P1 · TOPIC 2 8 marks

8.

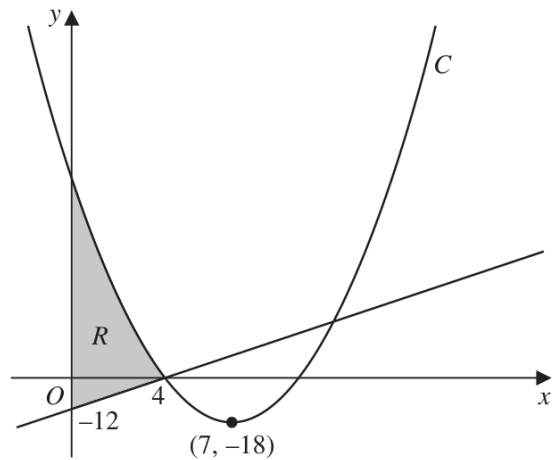


Figure 2

Figure 2 shows a sketch of the straight line  $l$  and the curve  $C$ .

Given that  $l$  cuts the  $y$ -axis at  $-12$  and cuts the  $x$ -axis at  $4$ , as shown in Figure 2,

- (a) find an equation for  $l$ , writing your answer in the form  $y = mx + c$ , where  $m$  and  $c$  are constants to be found.

(2)

Given that  $C$

- has equation  $y = f(x)$  where  $f(x)$  is a quadratic expression
- has a minimum point at  $(7, -18)$
- cuts the  $x$ -axis at  $4$  and at  $k$ , where  $k$  is a constant

- (b) deduce the value of  $k$ ,

(1)

- (c) find  $f(x)$ .

(3)

The region  $R$  is shown shaded in Figure 2.

- (d) Use inequalities to define  $R$ .

(2)

(Total for Question 8 is 8 marks)

**Q36** **PAST PAPER** Pearson IAL WMA11 · January 2024 · Q9**P1 · TOPIC 2 10 marks**

9. The curve  $C_1$  has equation  $y = f(x)$ .

Given that

- $f(x)$  is a quadratic expression
  - $C_1$  has a maximum turning point at  $(2, 20)$
  - $C_1$  passes through the origin
- (a) sketch a graph of  $C_1$  showing the coordinates of any points where  $C_1$  cuts the coordinate axes,

(2)

(b) find an expression for  $f(x)$ .

(3)

The curve  $C_2$  has equation  $y = x(x^2 - 4)$

Curve  $C_1$  and  $C_2$  meet at the origin, and at the points  $P$  and  $Q$

Given that the  $x$  coordinate of the point  $P$  is negative,

(c) using algebra and showing all stages of your working, find the coordinates of  $P$

(5)

(Total for Question 9 is 10 marks)

# ALL The Whole Topic on One Page

Tear this page out before the exam.

- 01 Solve** Calculator first. Clean → factorise, messy → formula.
- 02 Complete the square** Two-step structure. Turning point: flip inside, keep outside.
- 03 Discriminant**  $> 0$  two ·  $= 0$  one ·  $< 0$  none. Then a quadratic inequality in  $k$ .
- 04 Hidden quadratics**  $u = x^2, \sqrt{x}, 2^x$  ... reject impossible  $u$ .
- 05 Graphs & models** Shape, crossings, turning point. Regions: one inequality per edge.

**BOTTOM LINE** Roots → solve · turning point → complete the square · how many roots → discriminant.

# Answers

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Quadratics · for the teacher. Every answer checked with a computer algebra system.

## IDEA 01 · SOLVE

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- Q1** Dr Eslam · Calculator, then show it (a)  $x = 1, 5$  (b)  $x = \frac{-3 \pm \sqrt{41}}{4}$
- Q2** Practice · Easy (i)  $2x^2 + 5x - 12$  (ii)  $9x^2 - 16$  (iii)  $4x^2 + 4x + 1$
- Q3** Practice · Easy (i)  $1, -9$  (ii)  $\frac{1}{3}, 4$  (iii)  $\frac{3 \pm \sqrt{29}}{4}$
- Q4** Practice · Very Hard (a)  $(x + 3)^2$  (b)  $(x + 3y)^2$  (c)  $x = -3y$
- Q5** Pearson IAL WMA11 · June 2023 · Q2 (a)  $2x + 2y = 350$  (b)  $xy = 7350$  (c)  $x = 105, y = 70$

## IDEA 02 · COMPLETING THE SQUARE

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- Q6** Dr Eslam · Turning point  $(x - 3)^2 - 4; (3, -4)$
- Q7** Practice · Easy (i)  $(x+4)^2 - 20$  (ii)  $2(x+3)^2 - 23$  (iii)  $5\left(x - \frac{3}{10}\right)^2 + \frac{31}{20}$
- Q8** Practice · Hard (a)  $4(x+1)^2 - 9$  (b)  $(-1, -9)$  (c) roots  $-\frac{5}{2}, \frac{1}{2}$ ; y-int  $-5$
- Q9** Practice · Very Hard divide by  $a$ , complete the square, root
- Q10** Pearson IAL WMA11 · January 2022 · Q2 (a)  $13 - 2(x + 1)^2$  (b) max  $(-1, 13)$ , y-int  $11$  (c)  $x = -1$
- Q11** Pearson IAL WMA11 · June 2023 · Q3 (a)  $3(x + 2)^2 + 1$  (b) y-int  $13$ , min  $(-2, 1)$
- Q12** Pearson IAL WMA11 · October 2023 · Q11 (a)  $2(x - 3)^2 - 4$  (b)  $(3, -4)$  (c)  $y = -8x + 20$  (d)  $x \geq 0, y \geq 0, y \leq 20 - 8x, y \geq 2x^2 - 12x + 14$
- Q13** Pearson IAL WMA11 · October 2025 · Q5 (a)  $2(x - 4)^2 + 18$  (b)  $x = \frac{25}{4}$  (c)  $x \geq 0, y \geq \frac{9}{2}x, y \leq 2x^2 - 16x + 50$
- Q14** Pearson IAL WMA11 · June 2026 · Q5 (a)  $25 - 2(x - 3)^2$  (b)  $P(3, 25)$  (c)  $x \geq 0, y \geq \frac{25}{3}x, y \leq -2x^2 + 12x + 7$

### IDEA 03 · THE DISCRIMINANT

- Q15** Dr Eslam · Equal roots  $k^2 - 16 = 0 \Rightarrow k = \pm 4$
- Q16** Dr Eslam · No real roots  $16 - 4k < 0 \Rightarrow k > 4$
- Q17** Practice · Easy (i) -7 (ii) 40 (iii) 24
- Q18** Practice · Medium  $p^2 - 4q < 0$
- Q19** Practice · Very Hard (a)  $x = \frac{3k+1}{2(k+1)}$  (b)  $\left(\frac{5}{4}, -\frac{33}{8}\right)$
- Q20** Pearson IAL WMA11 · January 2023 · Q4  $0 < k < \frac{5}{9}$
- Q21** Pearson IAL WMA11 · October 2023 · Q6 (b)  $p < -\frac{2}{3}$  or  $p > 4$
- Q22** Pearson IAL WMA11 · June 2024 · Q4 (b)  $k = -13$ :  $P(1, -21)$
- Q23** Pearson IAL WMA11 · June 2025 · Q10 (a)  $x = -1, -\frac{1}{\sqrt[3]{7}}$  (b)  $k < 0$  or  $k > 5$

### IDEA 04 · HIDDEN QUADRATICS

- Q24** Dr Eslam · Substitute  $u = x^2$   $u = 1, 4$ :  $x = \pm 1, \pm 2$
- Q25** Practice · Medium  $x = \pm 2, \pm 3$
- Q26** Practice · Hard (a)  $x = 9$  (b)  $x = 8, -64$  (c)  $x = 2, 4$
- Q27** Practice · Very Hard (a)  $x = 16$  (b)  $x = 1, 2$
- Q28** Pearson IAL WMA11 · October 2019 · Q8  $x = 14 \pm 6\sqrt{5}$
- Q29** Pearson IAL WMA11 · January 2020 · Q5 (a)  $0, 3, -\frac{1}{2}$  (b)  $y = 6$  or  $y = -3$
- Q30** Pearson IAL WMA11 · October 2025 · Q4 (i)  $p = \frac{9}{4}$  (ii)  $x = \pm 2\sqrt{2}$

## IDEA 05 · GRAPHS AND MODELS

- Q31** Dr Eslam · Sketch cup; (0, 5), (1, 0), (5, 0); minimum (3, -4)
- Q32** Practice · Easy (a) -3; roots  $\frac{1}{2}$ , -3 (b) cup through (-3, 0),  $(\frac{1}{2}, 0)$ , (0, -3)  
roots  $-\frac{9}{4}$ ,  $\frac{8}{3}$ ; y-int -72
- Q33** Practice · Hard (a) 52 m (b) 56.5 m (c)  $3 + \sqrt{113} \approx 13.6$  s (d) 6 s
- Q34** Practice · Very Hard (a)  $y = 3x - 12$  (b)  $k = 10$  (c)  $2x^2 - 28x + 80$  (d)  $x \geq 0, y \geq 0, y \leq 2x^2 - 28x + 80$
- Q35** Pearson IAL WMA11 · January 2023 · Q8 (b)  $f(x) = 20x - 5x^2$  (c)  $P(-8, -480)$
- Q36** Pearson IAL WMA11 · January 2024 · Q9